

Muhammad Abdul Raffay

CONTACT INFORMATION	23 Years Old Computer Science Student Technology Enthusiast m.a.raffay.official@gmail.com +92 323 5073652 Rawalpindi, Pakistan GitHub: maraffayofficial-arch LinkedIn: muhammad-abdul-raffay
RESEARCH INTERESTS	Generative AI, deep learning for structured design generation, computer vision, and applied machine learning — currently pursuing Generative AI as the focus of my Final Year Project, centered on automated architectural plan generation.
PROFILE	Motivated BSCS Final year (8th semester) at Allama Iqbal Open University with hands-on experience in full stack MERN development, Currently transitioning focus toward Generative AI, with the Final Year Project centered on generative architectural design.
EDUCATION	Allama Iqbal Open University , Islamabad, Pakistan Bachelor of Science in Computer Science — 8th Semester (Ongoing) 2022 – Present IMCB I-10/1 , Islamabad, Pakistan Higher Secondary School Certificate (HSSC) Completed
TECHNICAL SKILLS	Languages: JavaScript (ES6+), Python, HTML, CSS Frameworks & Libraries: Node.js, Express.js, React.js, PyTorch Databases: MongoDB, MySQL APIs & Auth: RESTful APIs, JWT Authentication, CORS
PROJECTS	Floor Planner — Generative AI for Architectural Design — <i>Python, PyTorch</i> — GitHub - Built and completed a Generative AI model that automatically creates architecturally valid floor plans, learning spatial and structural constraints directly from data. - Designed the deep learning architecture in PyTorch and used OpenCV for spatial post-processing and plan validation. Attendance Prediction ML Project — GitHub - Designed an end-to-end time-series ML pipeline using Facebook Prophet, XGBoost and logistic regression integrated with a Flask REST API. - Connected to a MySQL database for persistent data storage; achieved a prediction accuracy of 0.93. Real-time Chat App — <i>MERN</i> — GitHub - Built a full-duplex real-time messaging application using WebSockets via Socket.IO. - Implemented room-based chat, user join/leave events, and live message broadcasting. Ecom Web App — <i>MERN</i> — GitHub - Delivered a full-stack e-commerce solution for a client covering UI, API, database design, and cloud deployment. - Handled ongoing maintenance, feature updates, and client-driven requirements over 1 year.
EXPERIENCE	Machine Learning Intern — Kaizen Hive 3 Months Gained hands-on experience with machine learning workflows, model development, and applied problem-solving in a professional setting.
CERTIFICATIONS	Neural Networks and Deep Learning — IBM (Coursera) 2025
LANGUAGES & INTERESTS	Languages: English (Fluent), Urdu (Fluent), Japanese (Novice) Interests: Reading, Chess, Content Creation, Programming, Machine Learning Research